AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 5 · STUDENT LESSON

Composing layers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Trace the layer. Name its output. Feed the next layer.

Compose a complete network while preserving dimensions,order and intermediate meanings.

Keep one habit: Main2→2→1flow → complete trace table → separate pictured detectors → shape/order probes → independent new network.

MAIN LESSON · SECTION 1

What does the next layer actually receive?

> Keep input,preactivation,hidden activation and output in separate boxes.

Learning goals

I can trace how one **layer** supplies the next **layer input**s.

I can check that connected **layer** dimensions and feature order agree.

I can explain why intermediate activations are not automatically class **prediction**s.

Retrieve the prior computation

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- For z = -2 and z = 3, what does ReLU return? Does this operation itself assign a class?

> **My starting explanation**

End goal

By the end, I can explain composing **layer**s using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Trace how one **layer** supplies the next **layer input**s

Check that connected **layer** dimensions and feature order agree

Explain why intermediate activations are not automatically class **prediction**s

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Connect two named transformations in the right order

> Both hidden units read the original input;the output reads their activations.

Complete hidden and output formulas

A network connects **layer**s so that the **output** vector of one **layer** becomes the **input** vector of the next. Use a fictional numerical **input** x = [x1, x2].

The first **layer** computes z1 = x1 - x2 and z2 = x1 + x2 - 2, then applies ReLU separately: h1 = max(0, z1), h2 = max(0, z2). Its **output** is the ordered vector h = [h1, h2].

The next **layer** computes y = 2h1 - h2 + 1. It receives h, not the original x. It uses weights [2, -1] and bias 1, with identity as its **output** activation.

This is a two-**layer** computation when counting the **layer**s with weights; an **input** placeholder is not counted here. All parameters are supplied, so this lesson performs no **training**.

Input/output sizes at each layer

Stage	Input size	Output size
First layer with ReLU	2 components	2 components
Output layer with identity	2 components	1 component

Distinguish parallel hidden units from the next layer

The two first-**layer** units both receive the same original x. They can be calculated in either order. Only after both activations are available do we evaluate the **output layer**.

Name given and missing facts

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Trace each intermediate value before the final output

A correct answer can hide an incorrect path;retain the complete trace.

Complete main model and worked input

Trace the evidence

STAGE	EVIDENCE	RESULT
Input x=[x1,x2]	Both units receive x	z=[x1−x2,x1+x2−2]
Two affine scores z	Apply ReLU to each	h=[h1,h2]
Ordered hidden pair h	Output weights[2,−1],bias1	One output[y]with identity

Take x = [3, 1]. First compute z1 = 3 - 1 = 2 and z2 = 3 + 1 - 2 = 2. ReLU gives h = [2, 2]. The **output layer** receives exactly this ordered pair and returns y = 2 times 2 - 2 + 1 = 3.

Record z, h and y separately so that a correct final number cannot hide an incorrect intermediate step.

All supplied trace rows

Main model input x · z1=x1−x2 · z2=x1+x2−2 · h=[ReLU(z1), ReLU(z2)] · y=2h1−h2+1

> **Main model input x:** [3, 1]
> **z1=x1−x2:** 2
> **z2=x1+x2−2:** 2
> **h=[ReLU(z1), ReLU(z2)]:** [2, 2]
> **y=2h1−h2+1:** 3

> **Main model input x:** [1, 3]
> **z1=x1−x2:** -2
> **z2=x1+x2−2:** 2
> **h=[ReLU(z1), ReLU(z2)]:** [0, 2]
> **y=2h1−h2+1:** -1

> **Main model input x:** [1, 1]
> **z1=x1−x2:** 0
> **z2=x1+x2−2:** 0
> **h=[ReLU(z1), ReLU(z2)]:** [0, 0]
> **y=2h1−h2+1:** 1

> **Main model input x:** [3, 0]
> **z1=x1−x2:** 3
> **z2=x1+x2−2:** 1

> **h=[ReLU(z1), ReLU(z2)]:** [3, 1]
>
> **y=2h1−h2+1:** 6

> **Main model input x:** [0, 1]
>
> **z1=x1−x2:** -1
>
> **z2=x1+x2−2:** -1
>
> **h=[ReLU(z1), ReLU(z2)]:** [0, 0]
>
> **y=2h1−h2+1:** 1

> **Main model input x:** [0, 2]
>
> **z1=x1−x2:** -2
>
> **z2=x1+x2−2:** 0
>
> **h=[ReLU(z1), ReLU(z2)]:** [0, 0]
>
> **y=2h1−h2+1:** 1

Runnable forward pass and its contract

```
def forward(x):
    if len(x) != 2:
        raise ValueError("Expected two input components")
    x1, x2 = x
    z = [x1 - x2, x1 + x2 - 2]
    h = [max(0, value) for value in z]
    y = 2 * h[0] - h[1] + 1
    return z, h, y

print(forward([3, 1]))  # ([2, 2], [2, 2], 3)
print(forward([1, 3]))  # ([-2, 2], [0, 2], -1)
```

The code assumes numeric **input**s with the agreed meaning. Its length check does not validate units or field order. Python indices 0 and 1 correspond to the first and second hidden components.

There is no feedback from y into h during this **forward calculation**.

Connect the picture and named values

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare a changed input while keeping the network fixed

> A revised layer size requires an explicit compatible definition.

Current dimension-mismatch task

- A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured second-input change

- Change: Change only v from 3 to1.

▼ **Check after predicting**

Prediction: h1 stays 2, h2 becomes 0, y becomes 2.

Reason: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See a changed hidden value flow into the next layer

> Only the second hidden computation changes in this supplied comparison.

Complete picture formulas and separate scope

Compose two hidden detectors before one **output** instead of skipping a **layer**.

For **input** [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. **Input** [4, 3].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The pictured model is a separate simpler network:h1=ReLU(u−2), h2=ReLU(v−1), y=h1−h2. At[4, 3], preactivations[2, 2]give h[2, 2], y0. Change only v3→1:preactivations[2, 0], h[2, 0], y2. Do not use the main model's formulas or output bias 1 here. The shared idea is that the output receives the computed hidden values, not the original input. Supplied detector names do not imply a learned biological mechanism.

Trace both hidden units and output

AI ACADEMY / AI-5/5 · 02 · TRACE THE CASE

Follow the deciding steps



01 First hidden unit

u-2=2, so h1=2.



02 Second hidden unit

v-1=2, so h2=2.



03 Compose output

y=2-2=0; the output uses activated hidden values, not raw inputs.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First hidden unit	u-2=2, so h1=2.
2. Second hidden unit	v-1=2, so h2=2.
3. Compose output	y=2-2=0; the output uses activated hidden values, not raw inputs.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Composition means one **layer**'s result becomes the next **layer**'s **input**. Compute both hidden units before combining them. At [4,3], their equal **output**s cancel.

Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**. The final subtraction must use activated hidden values rather than the raw measurements.

Predict the second-input change

- Change one thing: Change only v from 3 to1. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-5/5 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

y=2-2=0; the output uses activated hidden values, not raw inputs.

Changed condition

Change only v from 3 to1.

Worked changed-case result

h1 stays 2, h2 becomes 0, y becomes 2.
Why: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the deciding intermediate value and limits

- Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

- Read the labelled picture: What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main model[1,1]attempt

Complete each stage before using its output

AVAILABLE INPUT	COMPUTATION	VALUE PASSED ON
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Trace x = [1, 1]. Record z1, z2, h1, h2 and y. Explain why using x directly in the output formula would be a different model.

▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Composing layers

The facts we will use

AI ACADEMY / AI-5/5

Composing layers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First hidden unit

u-2=2, so h1=2.



2. Second hidden unit

v-1=2, so h2=2.



3. Compose output

y=2-2=0; the output uses activated hidden values, not raw inputs.

CHANGE ONE THING

Change only v from 3 to1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a shape or order bug before trusting the output

> Equal lengths do not guarantee matching meanings.

Complete dimension,order and activation comparisons

The **output layer** expects two **input**s because its two weights match h1 and h2. If the hidden **layer** is redesigned to return three activations, the **output layer** needs a compatible three-**input** definition.

Silently dropping an activation or inventing a third weight does not preserve the stated model.

Order matters even when lengths match. For x = [4, 1], h = [3, 3] happens to be unchanged by swapping, so that case cannot reveal an order bug. For x = [3, 0], h = [3, 1] gives y = 6.

Swapping only those hidden values gives 2 times 1 - 3 + 1 = 0. Swapping both the values and their matching weights would preserve the dot product.

Keeping the **output layer** fixed, replacing ReLU with identity changes the model. For x = [1, 3], identity gives h = [-2, 2] and y = -5; ReLU gives h = [0, 2] and y = -1.

More **layer**s or a different activation do not establish better **prediction**s without valid evaluation.

Choose an input that reveals the swapped-hidden bug

- Compare hidden vectors for x = [4, 1] and x = [3, 0]. Which **input** reveals a bug that swaps h1 and h2 but keeps **output** weights unchanged? Give correct and swapped **output**s for that **input**.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the forward path with less help

> Name the layer and value used at every step.

Current task and response

- For x = [1, 3], write both scores, both ReLU activations and y using the lesson model.

Trace x = [1, 1]. Record z1, z2, h1, h2 and y. Explain why using x directly in the output formula would be a different model.

Explain your prediction to a partner before comparing results.

Supported W2record

Use workbook W2 for the connected example “Composing **layer**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for Composing **layer**s. Proposed explanation: “The second hidden unit receives the first hidden unit **output**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured composition reasoning

- What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent new network

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Repair a dimension mismatch

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate shape,order,zero and information loss

> All probe inputs and expected intermediate checks are supplied.

Complete current task and runnable forward pass

A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

```
def forward(x):
    if len(x) != 2:
        raise ValueError("Expected two input components")
    x1, x2 = x
    z = [x1 - x2, x1 + x2 - 2]
    h = [max(0, value) for value in z]
    y = 2 * h[0] - h[1] + 1
    return z, h, y

print(forward([3, 1]))  # ([2, 2], [2, 2], 3)
print(forward([1, 3]))  # ([-2, 2], [0, 2], -1)
```

Complete normal,boundary,missing and bug-revealing cases

Complete main-network probe	Expected trace / action
Normal x[3, 1]	z[2, 2], h[2, 2], y3
Boundary x[1, 1]	z[0, 0], h[0, 0], y1; hidden zeros are not final class decisions
Changed input x[1, 3]	z[−2, 2], h[0, 2], y−1
Shape mismatch x[3, 1, 0]	ValueError:Expected two input components
Missing component x[3, None]	Outside numeric contract; request/validate before forward
Order-sensitive hidden h[3, 1]from x[3, 0]	Correct y6; swap only h to[1, 3]gives 0
Order-insensitive probe x[4, 1]	h[3, 3], y4; swap does not expose the bug
Representation collision x[0, 1]and[0, 2]	Both h[0, 0], y1; later fixed layers cannot distinguish them from h alone

Predict,run or trace,and explain

1. Predict the complete probes and then use the supplied forward function one case at a time. Keep z, h, y, actual output/error and an explanation of the first mismatch. A three-unit hidden redesign needs three output coefficients and an explicit bias or a documented selection operation; do not silently truncate. Input length is checked, but numeric validity and coordinate meanings are assumed.

   No target labels are supplied here, so a successful forward pass is not measured accuracy. Reusing raw x in the output skips a layer and defines a different model.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compose a different two-unit network

> Use the new formulas,record preactivations,and apply only its stated final class rule.

Complete new independent network

A new model takes [u, v]. Compute a = ReLU(u + v - 3), b = ReLU(u - v), then q = a + 2b - 1. Trace both **input**s [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when q >= 0.

State the hidden and **output** vector sizes.

First attempt and support record

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Compose a new network independently

Attempt this without hints

A new model takes [u, v]. Compute a = ReLU(u + v - 3), b = ReLU(u - v), then q = a + 2b - 1. Trace both **input**s [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when q >= 0.

State the hidden and **output** vector sizes.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the layer contract and full trace together

> The project record should identify which values each layer consumes.

Project artifact and evidence table

Add a labelled composing **layer**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace how one **layer** supplies the next **layer input**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Keep actual evidence and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a complete information path

> Explain what valid dimensions establish and what still needs labels.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Trace how one **layer** supplies the next **layer input**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Layer	A group of computations that maps an input vector to an output vector
Hidden activation	An intermediate value computed inside a network
Composition	Applying one function to the output of another
Forward calculation	Computing successive values from an input using fixed parameters

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For x = [1, 3], write both scores, both ReLU activations and y using the lesson model.
- Compare hidden vectors for x = [4, 1] and x = [3, 0]. Which **input** reveals a bug that swaps h1 and h2 but keeps **output** weights unchanged? Give correct and swapped **output**s for that **input**.
- Use the lesson model to verify h and y for [0, 1] and [0, 2]. Could a fixed later **layer** receiving only h distinguish the two original **input**s?

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent a/b/qnetwork for the first individual attempt.

Current explanation and response

- The computed y is a numerical **output**, not a verified probability or a class by itself. If we separately declare class 1 when y >= 0 and class 0 otherwise, the worked **input**s [3, 1] and [1, 3] receive classes 1 and 0.

  Their true labels are not supplied, so these calculations do not measure accuracy.

- A hidden value is a computed feature. We should not name h1 as confidence or h2 as danger without **evidence** for that interpretation.

  Distinct **input**s may also produce the same hidden vector: [0, 1] and [0, 2] both give h = [0, 0] and y = 1. A later deterministic **layer** seeing only h cannot recover the lost distinction.

- Next week uses longer forward-pass tables to locate the first incorrect intermediate value and check complete network calculations.

- Use the lesson model to verify h and y for [0, 1] and [0, 2]. Could a fixed later **layer** receiving only h distinguish the two original **input**s?

> **A team says its output is correct because all layers connect and its code runs. Which claim has it checked, and what evidence is still missing?**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Complete each stage before using its output

AVAILABLE INPUT	COMPUTATION	VALUE PASSED ON
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Compose two hidden detectors before one **output** instead of skipping a **layer**.

By the end, I can	How I will show it
Trace how one layer supplies the next layer inputs	Explain the deciding step, show a trace or test, and state one limit.
Check that connected layer dimensions and feature order agree	Explain the deciding step, show a trace or test, and state one limit.
Explain why intermediate activations are not automatically class predictions	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Compose two hidden detectors before one **output** instead of skipping a **layer**.
- Which labelled step matches this detail? y=2-2=0; the **output** uses activated hidden values, not raw **input**s.
- 1. Compose **output**
- 2. Second hidden unit
- 3. First hidden unit
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compose **output** y=2-2=0; the **output** uses activated hidden values, not raw **input**s. **Composition** means one **layer**'s result becomes the next **layer**'s **input**.

Compute both hidden units before combining them. At [4,3], their equal **output**s cancel. Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**.

The final subtraction must use activated hidden values rather than the raw measurements.

Changed-case question and optional hint

- Changed-case practice: Change only v from 3 to1. Which conclusion is supported by the supplied facts?
- 1. h1 stays 2, h2 becomes 0, y becomes 2.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: h1 stays 2, h2 becomes 0, y becomes 2. The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

Repair and retain the record

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network's final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dot products and weighted sums (ai-5/2). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

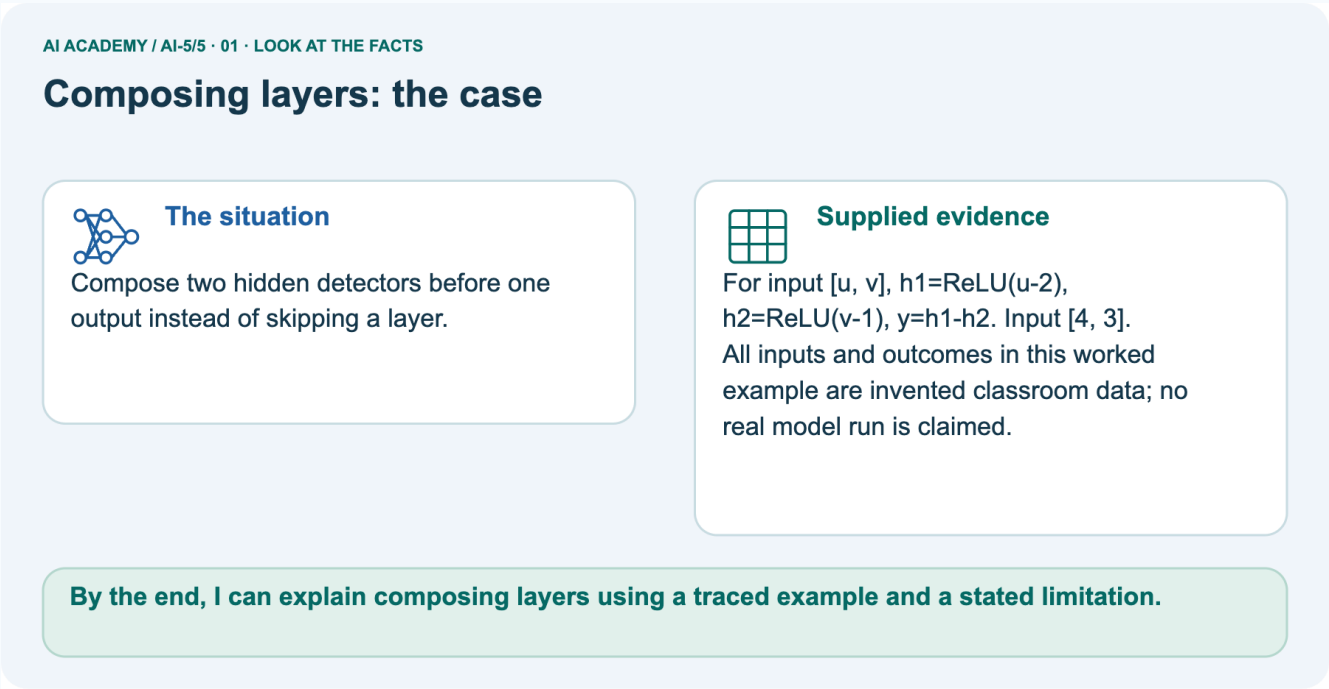
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Composing layers: the case



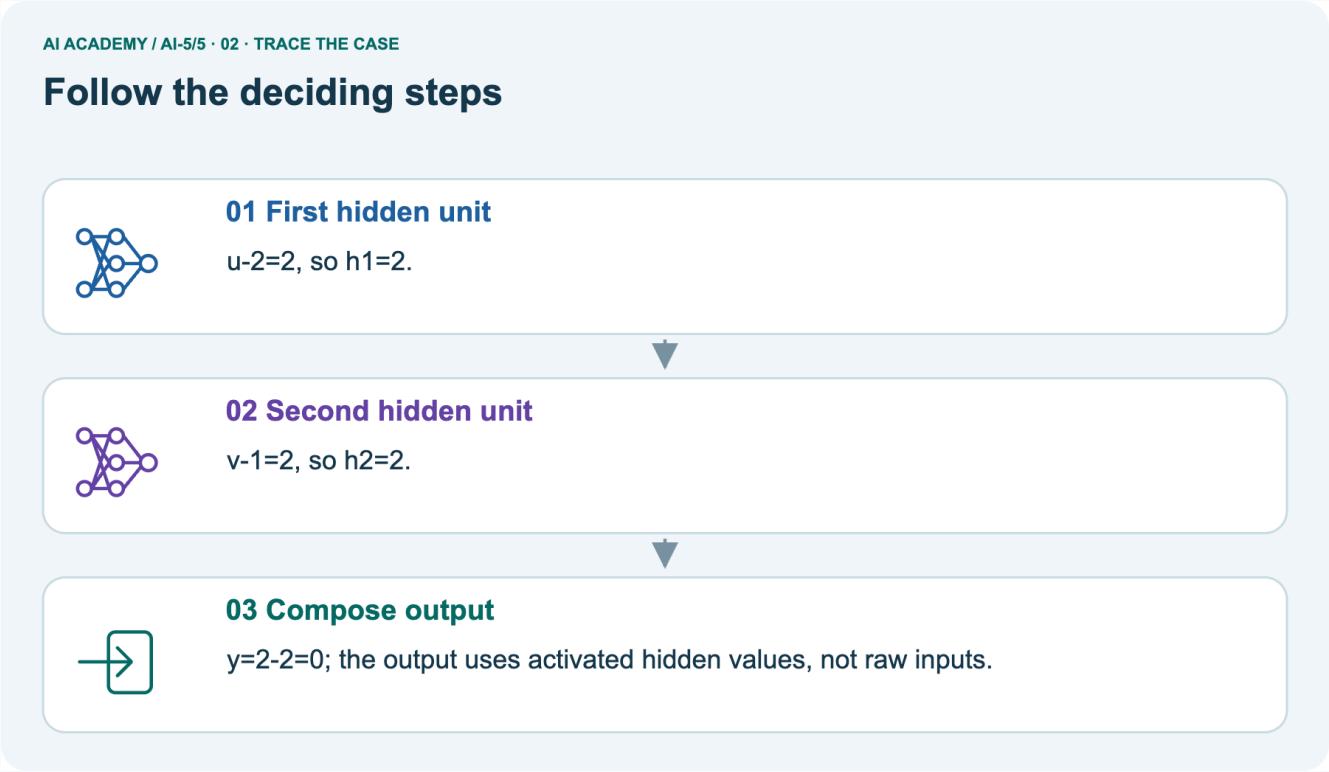
Follow the labelled picture with the matching explanation. Open the vector diagram

Compose two hidden detectors before one **output** instead of skipping a **layer**. Supplied facts: For **input** [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. **Input** [4, 3].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

First hidden unit: u-2=2, so h1=2. → Second hidden unit: v-1=2, so h2=2. → Compose **output**: y=2-2=0; the **output** uses activated hidden values, not raw **input**s.

Point to each numbered stage and explain which recorded **input** it uses.

03. Complete each stage before using its output

AI ACADEMY / AI-5/5 · 03 · INSPECT THE LESSON MODEL

Complete each stage before using its output

Available input	Computation	Value passed on
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

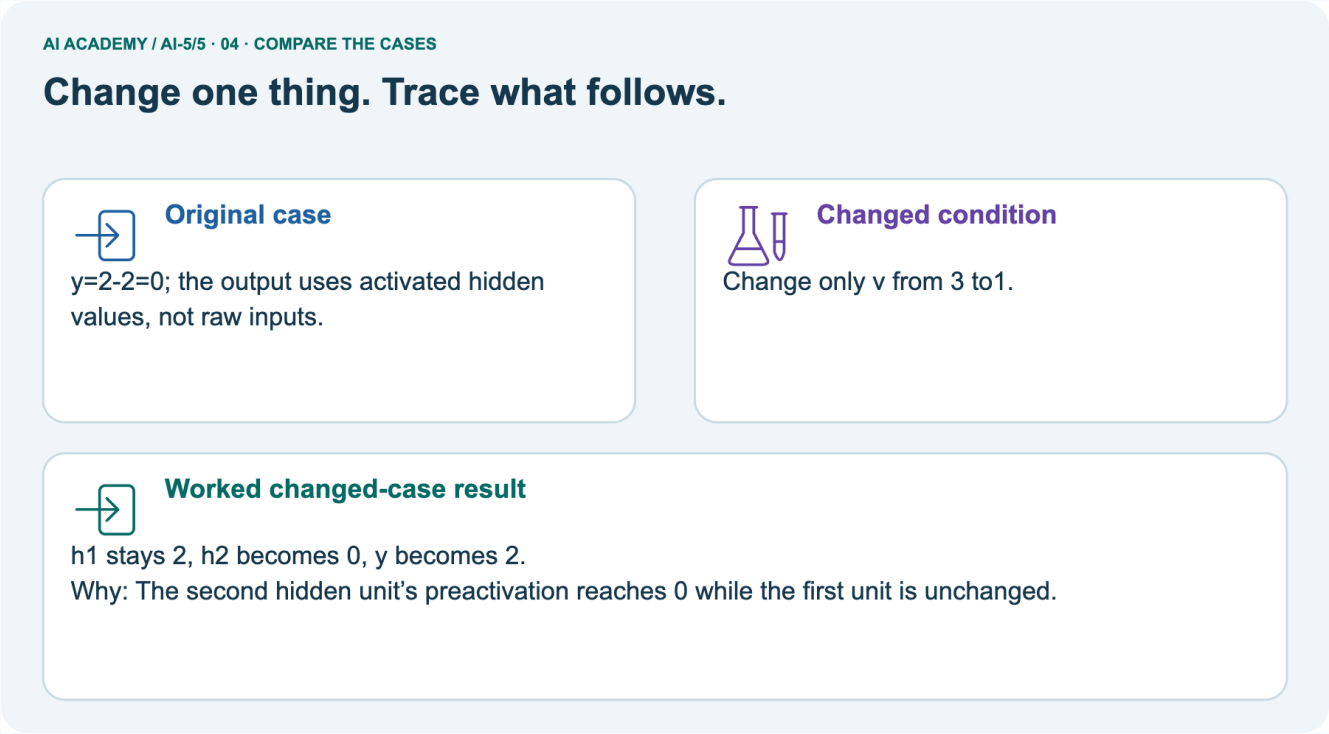
Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Follow the labelled picture with the matching explanation. Open the vector diagram

Complete each stage before using its **output**. x is [3, 1] | Both first **layer** scores | z is [2, 2]. z is [2, 2] | ReLU on each component | h is [2, 2]. h is [2, 2] | 2h1 - h2 + 1 | y is 3

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



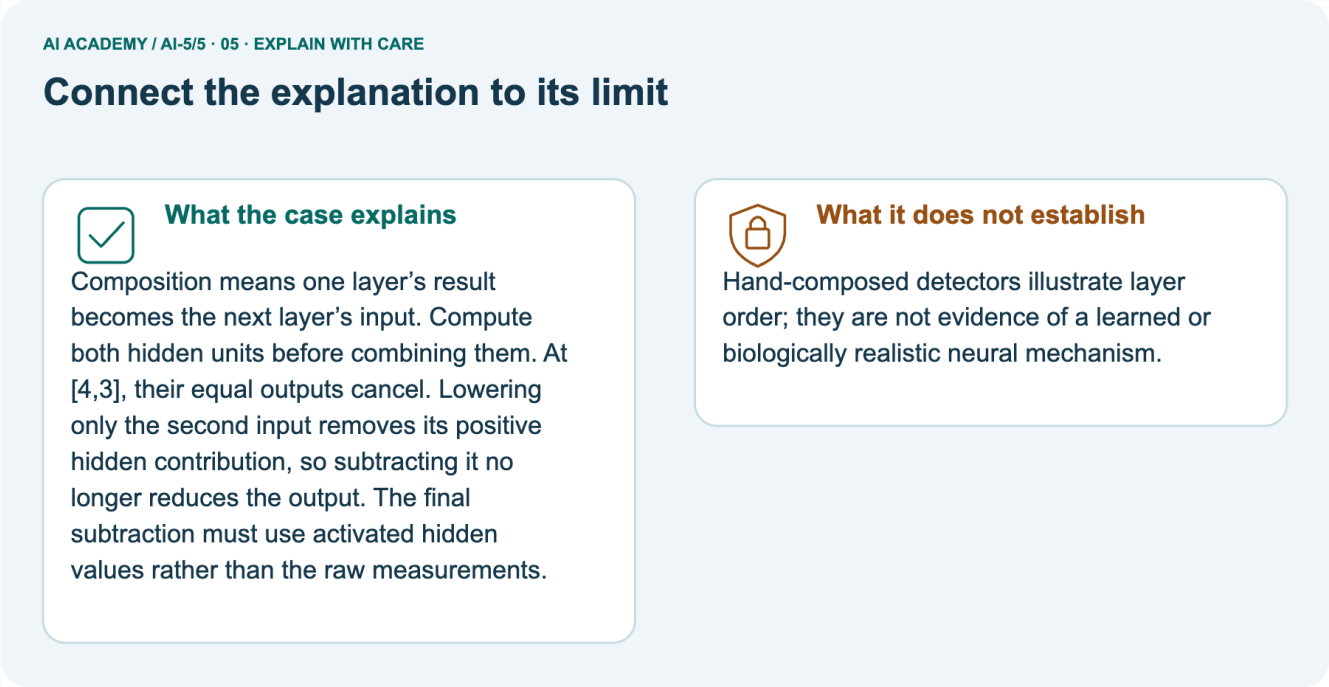
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: y=2-2=0; the **output** uses activated hidden values, not raw **input**s. Changed condition: Change only v from 3 to1. Expected result: h1 stays 2, h2 becomes 0, y becomes 2.

Reason: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

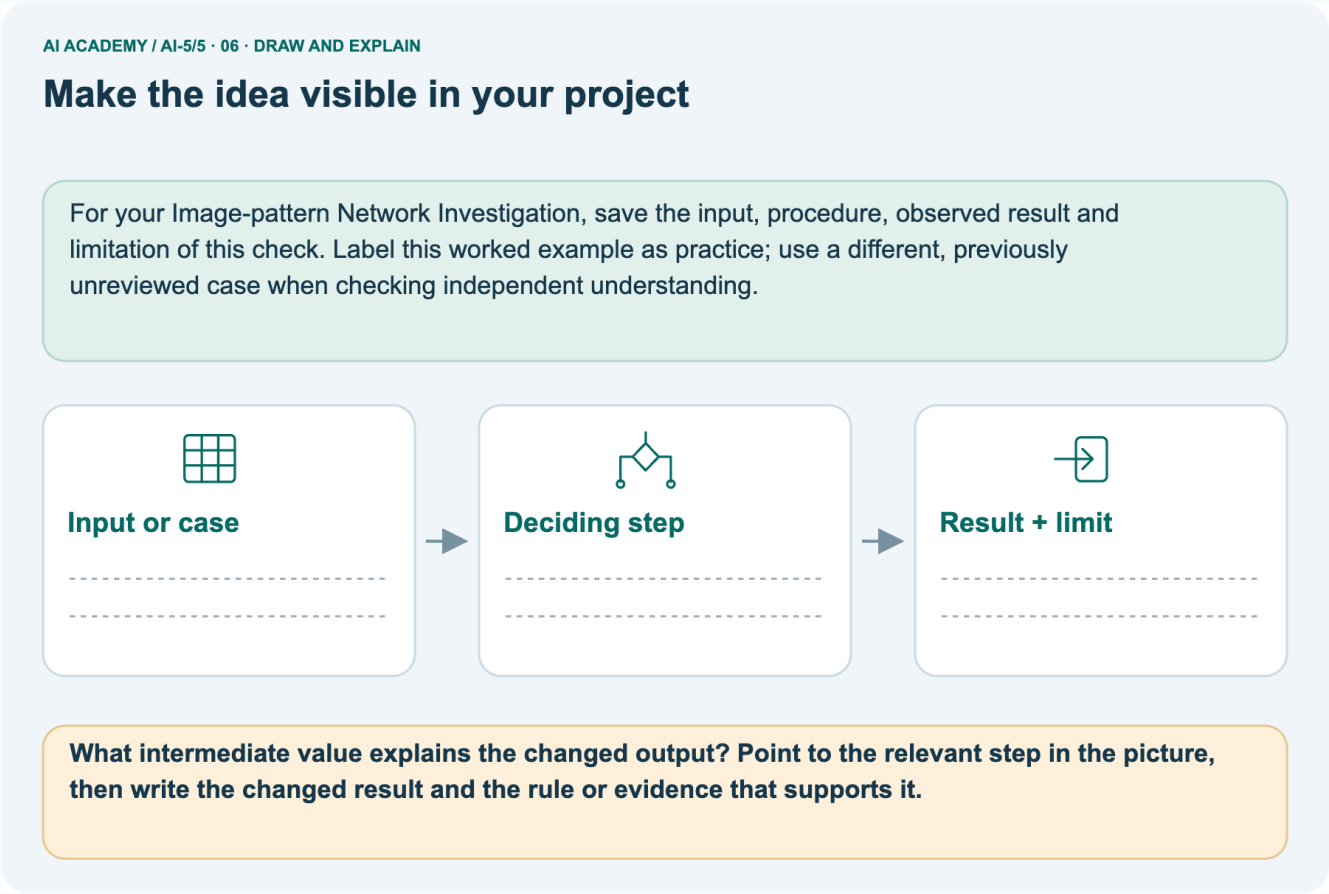
Composition means one **layer**'s result becomes the next **layer**'s **input**. Compute both hidden units before combining them. At [4,3], their equal **output**s cancel.

Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**. The final subtraction must use activated hidden values rather than the raw measurements.

Limit: Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Composing layers: the case

02. Follow the deciding steps

03. Complete each stage before using its output

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.